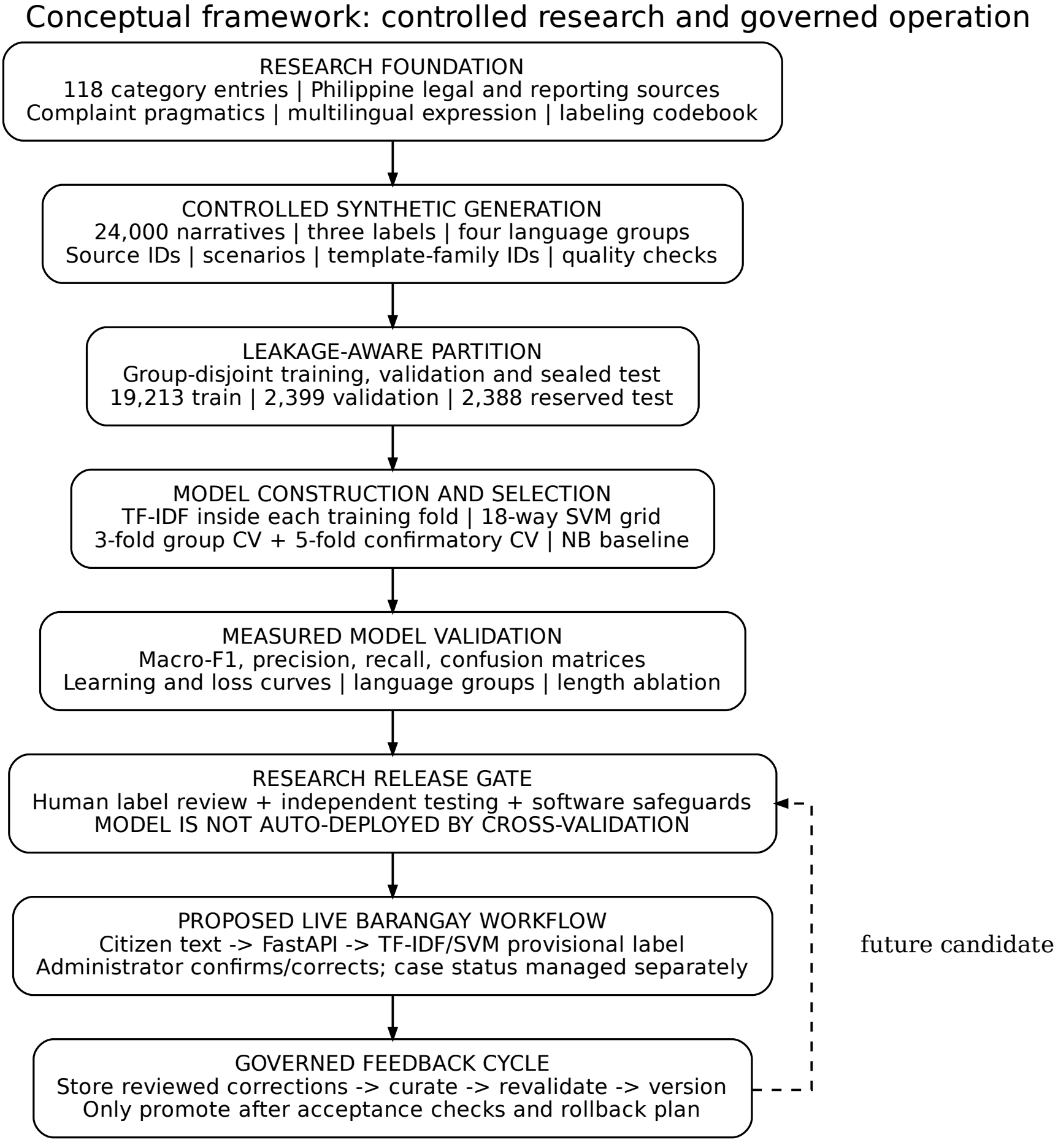
Conceptual framework: controlled research and governed operation
RESEARCH FOUNDATION
118 category entries | Philippine legal and reporting sources
Complaint pragmatics | multilingual expression | labeling codebook
CONTROLLED SYNTHETIC GENERATION
24,000 narratives | three labels | four language groups
Source IDs | scenarios | template-family IDs | quality checks
LEAKAGE-AWARE PARTITION
Group-disjoint training, validation and sealed test
19,213 train | 2,399 validation | 2,388 reserved test
MODEL CONSTRUCTION AND SELECTION
TF-IDF inside each training fold | 18-way SVM grid
3-fold group CV + 5-fold confirmatory CV | NB baseline
MEASURED MODEL VALIDATION
Macro-F1, precision, recall, confusion matrices
Learning and loss curves | language groups | length ablation
RESEARCH RELEASE GATE
Human label review + independent testing + software safeguards
MODEL IS NOT AUTO-DEPLOYED BY CROSS-VALIDATION
PROPOSED LIVE BARANGAY WORKFLOW
Citizen text -> FastAPI -> TF-IDF/SVM provisional label
Administrator confirms/corrects; case status managed separately
GOVERNED FEEDBACK CYCLE
Store reviewed corrections -> curate -> revalidate -> version
Only promote after acceptance checks and rollback plan
future candidate